Concept Design of a Rim-Driven Tidal Current Generator Integrated with a Classic SPAR Platform

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Based on the design of Team BluePower (four members), which received the First Prize at the 2026 Ocean Engineering CAE Competition (20 August 2026). The author led the team and performed the hydrodynamic and structural analyses.

Abstract. Deepwater SPAR platforms require several megawatts of electrical power for oil production, which is generated on board from fossil fuel. This study presents the concept design of an auxiliary tidal current power system integrated with a classic SPAR. Because the hull interior is occupied by risers and process equipment, a shaftless rim-driven generator is placed around the lower hull and driven by a three-bladed H-Darrieus rotor with a radius of 24 m and a blade height of 40 m. The blade section and operating point were selected from double-multiple-streamtube (DMST) analyses of four NACA sections in QBlade. The blade shell and support arms were sized for the 100-year current of 2.37 m/s, the support-arm section was obtained by constrained mass minimization, and both components were designed for neutral buoyancy so that the rotor imposes no net weight on the platform. Including arm drag and generator losses, the system produces 1.09 MW at the rated current of 1.5 m/s, approximately 9.4% of the estimated platform demand of 11.6 MW.

Keywords: SPAR platform; tidal current energy; vertical-axis turbine; rim-driven generator; DMST; neutral buoyancy

1. Introduction

Floating production platforms are designed to withstand environmental loads from wind, waves, current and ice. In design practice these loads are treated as hazards to be resisted rather than as resources. At the same time, a production platform must meet its own electrical demand, which on a deepwater SPAR amounts to several megawatts and is supplied by on-board generation from fossil fuel. This study examines whether one of the environmental loads, the ocean current, can be converted into electrical power on the platform itself.

Vertical-axis current turbines are suited to this application because they accept flow from any direction without a yaw mechanism and operate on lift with high power coefficients; their hydrodynamic behaviour has been investigated numerically and experimentally by Coiro et al. [5]. Integration with a SPAR, however, is constrained by the hull: risers and production equipment occupy its centre, so a turbine shaft entering the hull would interfere with them. Rim-driven machines avoid the shaft by placing the electrical generator at the periphery of the rotor, and their coupled electromagnetic and hydrodynamic behaviour in tidal applications has been modelled by Drouen et al. [10].

The present study combines these two elements. An H-Darrieus rotor is mounted on a ring that rotates around the lower hull of a classic SPAR, and the ring forms the rotor of a slotless rim generator. Section 2 defines the design basis. Section 3 describes the system configuration. Section 4 presents the hydrodynamic analysis used to select the blade section and rated operating point. Section 5 covers the structural design of the blades and support arms, including their neutral buoyancy. Section 6 estimates the net power output, and Sections 7 and 8 discuss the limitations of the design and summarize the findings.

2. Design basis

The host platform is modelled on the Tubular Bells classic spar (hull length 176 m, diameter 26 m). Its rated electrical demand was estimated at 11.6 MW for a peak oil production of 60,000 barrels per day, using the American Petroleum Institute standards for compressors, pumps and separators [1–3]. The metocean conditions correspond to a deepwater site east of Trinidad [4] and are listed in Table 1.

Table 1. Design conditions.

Parameter	Value	Remarks
Water depth	2,000–4,000 m	Deep water east of Trinidad [4]
Seawater density, ρ	1,022.76 kg/m^3	[4]
Rated current, V	1.5 m/s	Normal operation
Cut-out current	2.0 m/s	Turbine stopped above this speed
Extreme current	2.37 m/s	100-year return period; ultimate limit state
Target output	1 MW class	At the design tip-speed ratio

The coupled motion of the rotor and the SPAR, fabrication, and offshore installation are outside the scope of this study.

3. System configuration

The general arrangement is shown in Fig. 1 and the principal particulars in Table 2. Three H-Darrieus blades are carried by six support arms on a ring that rotates around a 20 m section of the lower hull. The ring holds the permanent magnets and back iron of a slotless generator, and the stator coils, sealed in epoxy against seawater ingress, are mounted on the hull across the air gap. The aspect ratio of the rotor (blade height to rotor diameter, 0.833) follows that of the Kobold turbine [5].

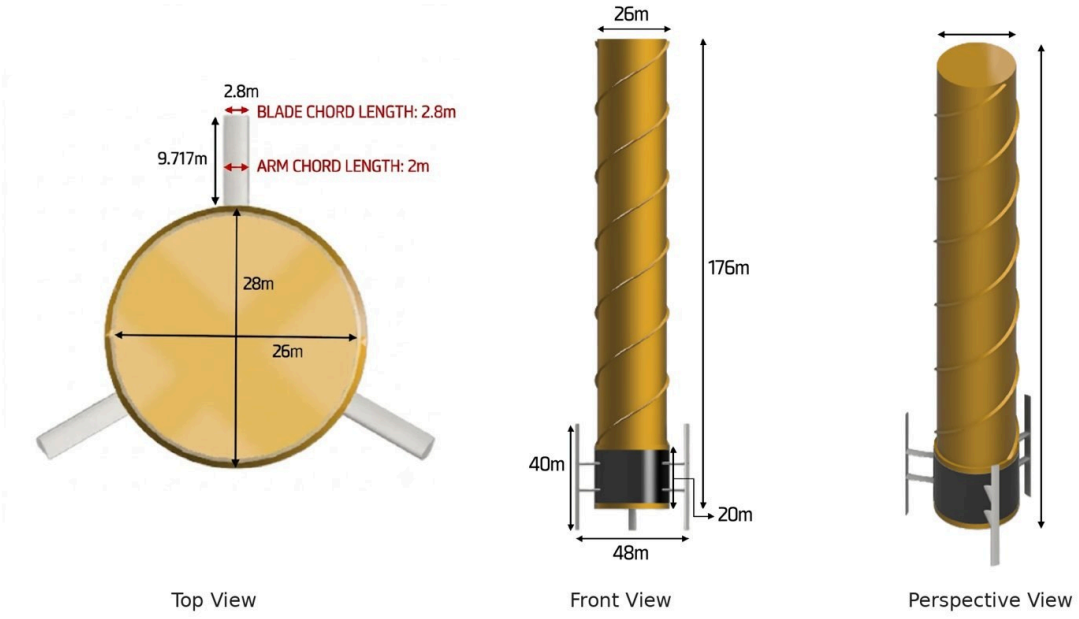


Fig. 1. General arrangement of the system: top, front and perspective views. Dimensions in metres.

Table 2. Principal particulars.

Component	Particulars
Rotor	H-Darrieus, 3 blades, NACA 0018, chord 2.8 m, height 40 m, radius 24 m
Support arms	6 arms, length 9.717 m; S690 circular tube in a NACA 0040 GFRP fairing of chord 2.0 m
Generator	Slotless rim-driven permanent-magnet machine on a 20 m rotating section
Host hull	Classic spar, length 176 m, diameter 26 m

4. Hydrodynamic analysis

4.1 Method

Rotor performance was predicted with the double-multiple-streamtube (DMST) model implemented in QBlade. The model divides the rotor into a set of streamtubes and solves the upstream and downstream halves of each separately, so that the velocity deficit across the rotor is accounted for at a small fraction of the cost of a CFD analysis. This makes it practical for comparing a large number of geometries in the early design stage. Because the relative velocity seen by a blade varies over each revolution, polars at a single Reynolds number would misrepresent the lift and drag; multi-Reynolds polars covering the operating range (Re ≈ 4×10^6 to 4×10^7) were used instead.

The hydrodynamic power available to the rotor is

$$P_{\text{avail}} = \tfrac{1}{2}\,\rho A V^3, \qquad A = 2RH \tag{1}$$

where R is the rotor radius and H the blade height. With A = 1,920 m², the available power at the rated current is 3.31 MW.

4.2 Section selection

Four symmetric sections (NACA 0012, 0015, 0018 and 0021) were analysed over a solidity of 0.10–0.50 and a tip-speed ratio of 0–8, both in steps of 0.05, where the tip-speed ratio is

$$\lambda = \frac{\omega R}{V}. \tag{2}$$

The best operating point of each section is given in Table 3 and Fig. 2. Thinner sections give slightly higher power coefficients, but the differences are small: NACA 0018 is 1.8% below NACA 0012 while being 50% thicker, which increases the section modulus available to resist bending of the 40 m blade. NACA 0021 loses more than 3%. NACA 0018 at a solidity of 0.35 was therefore selected, which corresponds to a chord of 2.8 m for three blades.

Table 3. Best operating point of each section from the DMST sweep.

Section	Solidity	λ	$C_{p,max}$	Relative to NACA 0012
NACA 0012	0.30	4.0	0.5490	–
NACA 0015	0.30	4.0	0.5442	−0.9%
NACA 0018	0.35	3.5	0.5392	−1.8%
NACA 0021	0.35	3.5	0.5318	−3.1%

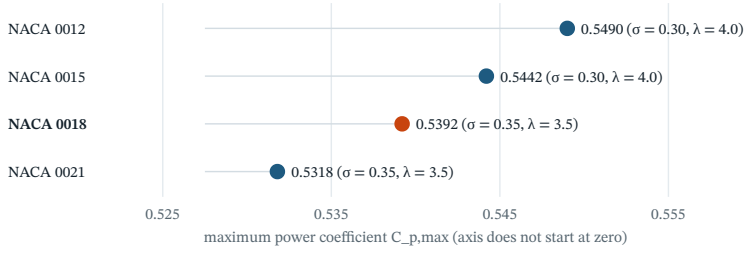


Fig. 2. Maximum power coefficient of each section. The selected section is shown in orange.

4.3 Rated operating point

At C_p = 0.5392 the rotor extracts 1.787 MW. The rated rotational speed is 2.15 rpm (ω = 0.225 rad/s), which corresponds to λ ≈ 3.6 and lies on the flat top of the C_p–λ curve next to its maximum at λ = 3.5. The rotor torque at the rated point follows from

$$T = \frac{P}{\omega} = \frac{1.787 \times 10^6}{0.225} = 7.93 \times 10^6\ \text{N m}. \tag{3}$$

5. Structural design

5.1 Ultimate limit state loads

The ultimate limit state was evaluated at the 100-year current of 2.37 m/s. In normal operation the rotor is stopped above the cut-out speed of 2.0 m/s, but the design must also withstand a failure in which the generator and brake lose their holding torque and the rotor runs free. Without any resisting torque, the rotor accelerates until the hydrodynamic driving torque falls to zero; at 2.37 m/s this occurs at $\lambda = 7.25$, the runaway tip-speed ratio, and no higher rotational speed can be reached. The tip-speed ratio was therefore varied from 0 (parked) to 7.25 (runaway) in steps of 0.05, and the normal and tangential distributed loads, q_n and q_t, were evaluated at the mid-span section of the blade. As shown in Fig. 3, q_n increases monotonically with λ and reaches 179.4 kN/m at $\lambda = 7.25$ and an azimuth angle of 42.5°, while q_t peaks near $\lambda = 2.7$ and falls to 7.39 kN/m at $\lambda = 7.25$. In the parked condition q_n is only about 13 kN/m, so the runaway condition governs, and the loads at $\lambda = 7.25$ were adopted for design.

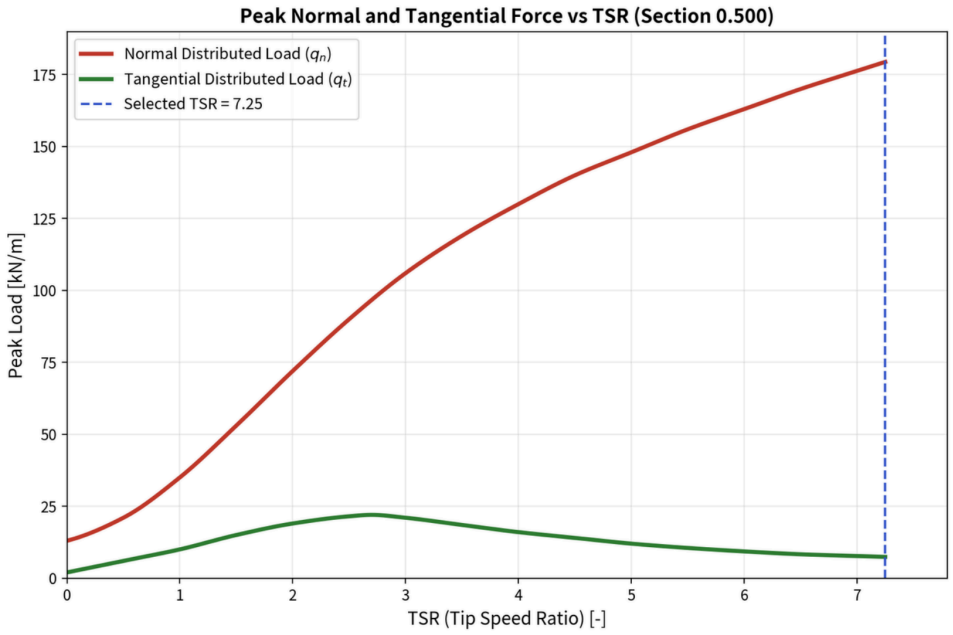


Fig. 3. Peak normal and tangential distributed loads at blade mid-span versus tip-speed ratio, V = 2.37 m/s.

5.2 Blade shell

Each blade is supported by two arms and overhangs them by a = 10 m. Treating the overhang as a cantilever under the peak normal load gives the governing flapwise bending moment

$$M_n = \frac{q_n a^2}{2} = \frac{179.4 \times 10^2}{2} = 8{,}970 \ \text{kN m}, \tag{4}$$

whereas the edgewise moment is 369.5 kN m, giving a combined moment of 8,978 kN m. The blade is made of glass-fibre-reinforced polymer (GFRP) with a tensile strength of 1,100 MPa and a compressive strength of 675 MPa [7]. Applying a safety factor of 2.46 based on DNV-OS-C501 [6] to the compressive strength gives

$$\sigma_{\text{allow}} = \frac{675}{2.46} = 274.4 \ \text{MPa}. \tag{5}$$

Numerical integration over the NACA 0018 contour gives the flapwise section modulus of a thin shell of thickness t as $Z = 0.786 t$ (Z in m³, t in m). The bending condition $M_n / Z \le \sigma_{\text{allow}}$ then requires

$$t \ge \frac{M_n}{0.786 \, \sigma_{\text{allow}}} = 41.6 \ \text{mm}. \tag{6}$$

A shell thickness of 43.6 mm was adopted. The calculation assumes that the shell alone carries the bending moment. The spar caps and shear webs shown in Fig. 4 follow the box-girder topology of Jeong and Yang [7], and their dimensions were determined by the neutral-buoyancy condition of Section 5.5; their contribution to bending stiffness is not included, so the strength estimate is conservative.

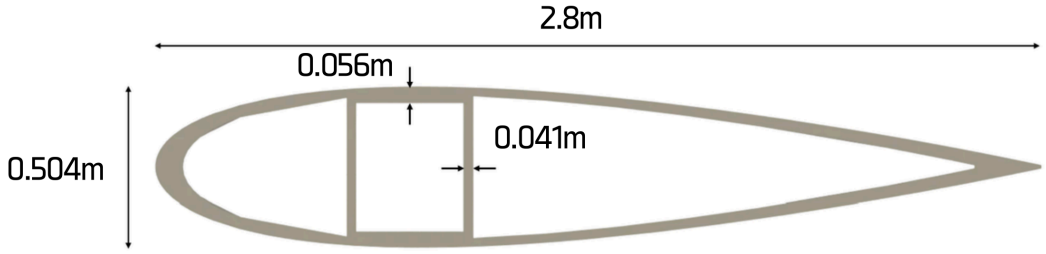


Fig. 4. Blade cross-section (NACA 0018, chord 2.8 m): GFRP shell (43.6 mm) with a box spar formed by two spar caps (56 mm) and two shear webs (41 mm).

5.3 Support arm loads

The blade loads are transmitted through the blade–arm joint to the root of the arm at the rotating ring, where all load components reach their maximum (Fig. 5). The axial force is the sum of the centrifugal force of the blade and the reaction of the normal load over half the blade length, $L/2 = 20$ m. The normal load over the overhang produces bending in the vertical plane, the tangential load produces bending in the horizontal plane over an effective length $L_{eff} = 10$ m, and the edgewise moment of the blade produces torsion. The resulting design loads are listed in Table 4.

Table 4. Design loads at the arm root.

Load	Expression	Value
Axial force	$m_b R\omega^2 + q_n L/2 = 392 + 3{,}588$ kN	3,980 kN
Vertical bending moment	$q_n a^2/2$	8,970 kN m
Horizontal bending moment	$(q_t L/2)L_{eff} = 147.8$ kN $\times$ 10 m	1,478 kN m
Combined bending moment	$(M_v^2 + M_h^2)^{1/2}$	9,091 kN m
Torsional moment	$q_t a^2/2$	369.5 kN m

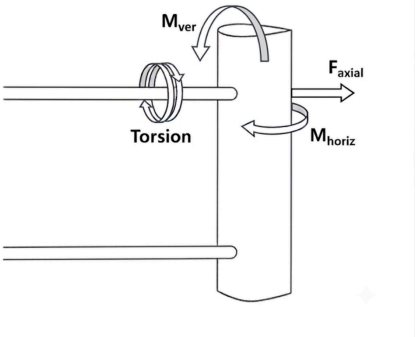


Fig. 5. Load components at the arm root.

5.4 Support arm section

Each arm consists of an S690 steel tube (yield strength 690 MPa) enclosed in a GFRP fairing with a NACA 0040 profile, which reduces the drag of what would otherwise be a bluff body. The outer diameter D, wall thickness t and fairing thickness ratio x were obtained from the following problem:

$$\min_{D,\,t,\,x} \; m(D,t) = \pi\,(D-t)\,t\,\rho_s L_{\text{arm}} \tag{7}$$

subject to

$$\begin{aligned} &\sigma_{\text{vm}} = \sqrt{\sigma^2 + 3\tau^2} \le \sigma_{\text{allow}}, \qquad |W(D,t,x) - B(x)| = 0, \\ &D \le c\,x - 2t_{\text{skin}} - \delta, \qquad x \le 0.40, \end{aligned} \tag{8}$$

where ρ_s = 7,850 kg/m³, L_{arm} = 9.717 m, σ is the normal stress from the axial force and combined bending moment, τ is the shear stress from torsion, W and B are the weight and buoyancy of the faired arm, c = 2.0 m is the fairing chord, t_{skin} = 10 mm, and δ is the assembly clearance. The limit x ≤ 0.40 prevents flow separation on the fairing. The allowable stress was determined with the load and resistance factor design method of DNV-OS-C101 [8], using a load factor of 1.2 and a material factor of 1.15, which gives σ_{allow} = 690/1.38 = 500 MPa.

Table 5. Optimal support arm section.

Quantity	Governing condition	Result
Outer diameter, D	Fit inside fairing (active)	778.5 mm
Wall thickness, t	Neutral buoyancy (active)	58.0 mm
Fairing profile	x ≤ 0.40 (active)	NACA 0040
Von Mises stress	≤ 500 MPa	444.3 MPa

At the optimum, the geometric, buoyancy and separation constraints are active, while the stress constraint is satisfied with a margin of 11%. The section is therefore governed by space and buoyancy rather than by strength. It would also satisfy a more conservative combined factor of about 1.5, for which the allowable stress is 460 MPa.

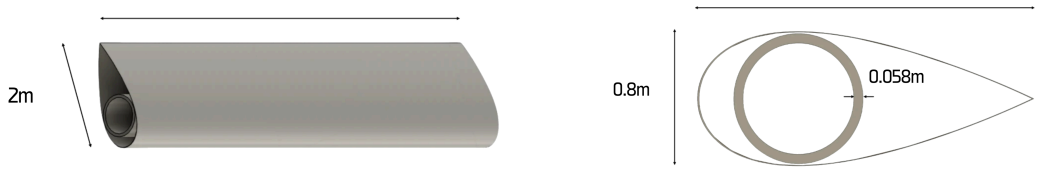


Fig. 6. Support arm: faired arm segment (left) and cross-section of the S690 tube (wall 58 mm) inside the NACA 0040 fairing of chord 2.0 m and thickness 0.8 m (right).

5.5 Neutral buoyancy

Because the payload capacity of a SPAR is limited, the blades and arms were designed so that each displaces its own mass and the rotor adds no net weight to the platform. Table 6 summarizes the balance for one blade and one arm. For the blade, the volume of GFRP required to match the displacement of the 40 m blade fixed the dimensions of the spar caps and shear webs; for the arm, it fixed the wall thickness of the steel tube.

Table 6. Mass and displacement of one blade and one arm (ρ_{GFRP} = 2,000 kg/m³, ρ_s = 7,850 kg/m³, ρ = 1,022.76 kg/m³).

Component	Material	Mass (kg)	Displaced volume (m³)	Displacement (kg)
Blade	GFRP shell 10.18 m³, spar caps 1.55 m³, shear webs 1.36 m³	26,180	25.60	26,183
Arm	Steel tube 10,012 kg, GFRP fairing 881 kg	10,893	10.651	10,893

6. Power output

Two losses reduce the power extracted by the blades. The first is the drag of the support arms, which move through the water at a relative speed of 5.81 m/s (Re = 1.26×10^7). With a drag coefficient of 0.0793 for the

faired section [9], the total arm drag is 66.0 kN, which produces a loss torque T_{arm} of 1.19×10^6 N m and a power loss of 0.27 MW. The second is the combined electromagnetic loss of the generator and the friction loss of seawater in the air gap. It was represented by a factor $\eta_g = 0.72$, an assumed value based on the loss trends reported for a comparable shaftless rim-driven tidal generator [10]. The net output is

$$P_{\text{net}} = \eta_g \left(T - T_{\text{arm}}\right) \omega = 0.72 \times 6.74 \times 10^6 \times 0.225 = 1.09\ \text{MW}. \tag{9}$$

Table 7 and Fig. 7 show the reduction from the rotor to the net output. The net output corresponds to an overall power coefficient of 0.33 and covers approximately 9.4% of the platform demand of 11.6 MW.

Table 7. Power coefficient, torque and power from the rotor to the net output (ω = 0.225 rad/s).

Stage	C_p	Torque (N m)	Power (MW)
Rotor	0.5392	7,927,035	1.787
After arm drag	0.4584	6,739,156	1.519
After generator and gap friction losses	0.330	4,851,486	1.092

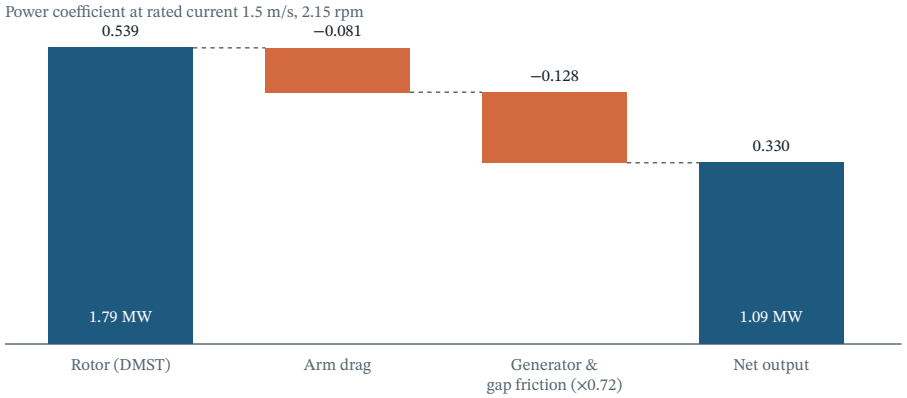


Fig. 7. Reduction of the power coefficient from the rotor to the net output.

7. Discussion

The analyses in this study are at the concept level (technology readiness level 2–3). Rotor performance was obtained from DMST and the structures were sized with beam theory, so blade and arm stresses should be verified by CFD and finite element analysis at the next stage, and the coupled motion of the rotor and the SPAR has not yet been examined.

Three structural checks that are normally required for a tidal turbine were not carried out. The blades undergo one load cycle per revolution, about 3,100 cycles per day at the rated speed, so fatigue of the GFRP laminate and of the blade–arm joints is likely to govern the detailed design rather than the extreme load. The compressive side of the 43.6 mm shell, spanning a 2.8 m chord, must also be checked for local buckling, which may require the spar caps and shear webs to be sized for strength rather than for buoyancy alone. Finally, the bonded and bolted connections between blade, arm and ring were not designed.

The interaction between the rotor and the platform was likewise not evaluated. The rotor thrust adds to the current load on the hull and its mooring, the generator ring adds mass near the keel, and the rotating system may change the motion response of the SPAR. These effects would be the first to examine in a feasibility study carried out with the platform designer.

The largest single uncertainty in the power estimate is the generator and gap-friction factor of 0.72, which was taken from the literature rather than calculated for this machine. A change of 0.1 in this factor changes the net output by about 0.15 MW.

Two further points require attention in detailed design. First, the steel tube of the support arm fills its fairing with no remaining clearance, so a fabrication tolerance must be restored at the expense of either wall thickness or buoyancy. Second, the centrifugal force used for the arm, 392 kN, back-calculates to $\lambda \approx 8$

at the extreme current and was carried over from an earlier load case; at the runaway value $\lambda = 7.25$ it is 322 kN. Because the arm stress is dominated by bending (about 413 of 444 MPa), the correction changes the von Mises stress by less than 1 MPa.

8. Conclusions

A shaftless tidal current generator integrated with a classic SPAR was designed at the concept level. The NACA 0018 H-Darrieus rotor was selected from a DMST parametric study as a compromise between power coefficient and structural thickness, and it extracts 1.79 MW at the rated current of 1.5 m/s. The blade shell requires a thickness of at least 41.6 mm under the 100-year current, and the optimal support arm is governed by geometric and buoyancy constraints rather than by stress. Both components are neutrally buoyant. After arm drag and generator losses, the system delivers 1.09 MW, approximately 9.4% of the platform demand, which indicates that environmental loads on an offshore platform can be partly converted into a useful power supply.

Acknowledgements

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